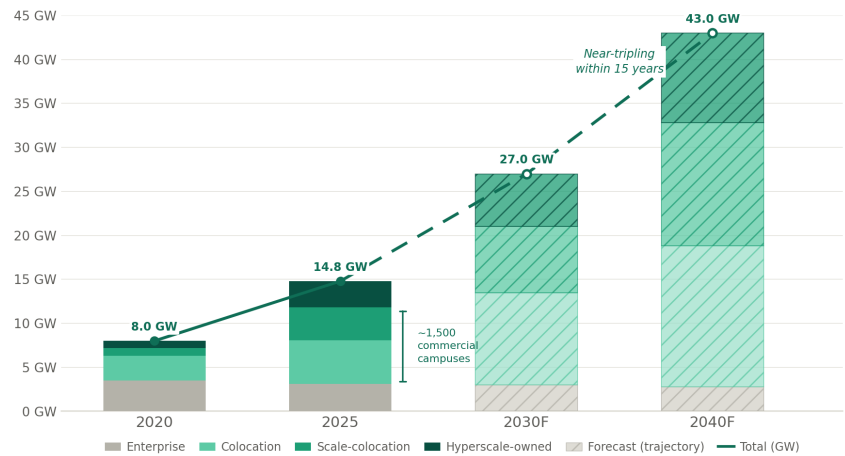


Why this matters now

Europe's data-centre sector has entered a phase in which digital demand is growing faster than the grid, permitting, equipment and operating-evidence frameworks within which capacity has to be developed, connected and operated. What were once medium-sized facilities serving enterprise and cloud customers have become large industrial loads whose connection geometry, cooling architecture, heat-reuse obligations and evidence requirements place them inside the same infrastructure logic as conventional power-system assets.

The scale of this shift is striking. Installed IT power across the European data-centre landscape reached approximately 14.8 GW by end of 2025, carried across more than 10,500 facilities. The strategic system question, however, is shaped by a much smaller group: roughly 1,500 commercial colocation, scale-colocation and hyperscale-owned campuses that carry the growth dynamic and the majority of connection, permitting and operating pressure. Forecasts converge on a range of 25–30 GW by 2030 and point to over 40 GW on a continued growth trajectory beyond 2040 — a near-tripling of connected European data-centre capacity within fifteen years, placing this infrastructure class in the order of large national power-system portfolios.



Source: EUDCA / Pb7 State of European Data Centres 2026; ENTSO-E May 2026; CBRE Research Q1 2026. 2030F and 2040F are rounded infrastructure-scale corridors. Note: x-axis intervals are not equal in time (5 yr / 5 yr / 10 yr). Annualised growth: +1.4 GW/yr (2020–25) → +2.4 GW/yr (2025–30) → +1.6 GW/yr (2030–40).

Germany holds the largest installed IT capacity in Europe, supported by economic scale, the Frankfurt interconnection ecosystem and a specific regulatory density. Connection requests from data centres, renewable generators and new industrial loads have begun to exceed available capacity, engineering bandwidth and system-planning certainty at both transmission and distribution level. The recently introduced maturity procedure for transmission-grid connection requests — the Reifegradverfahren — marks the moment at which connection capacity begins to be allocated by demonstrable project quality rather than by request order. It is the formal signal that the data centre has become a grid-infrastructure question, not merely a real estate one.

From digital service to utility infrastructure

The decisive engineering object is therefore not the server room, the rack, the building or the grid-connection package in isolation. It is the bounded campus as a coupled whole: electrical architecture, thermal management, compute and workload layer, connectivity, security, data and evidence systems, sustainability interfaces, and commercial and governance structure, all carried together. The quality of the asset does not arise from the strength of individual components. It arises from the coherence of these layers across the project, the connection regime and decades of operating life.

AI demand amplifies every physical dimension of this challenge. It increases rack-level power density, shifts cooling from air to liquid, strengthens the need for high-bandwidth network fabrics, sharpens commissioning obligations and reduces tolerance for supply-chain disruption. A campus that manages this transition while remaining technically operable, evidentially legible, institutionally accountable and commercially usable is not merely well-built. It is governable.

Grid connection as the upstream quality question

Grid connection has become a scarce infrastructural entry resource. Across Europe and in Germany, connection requests increasingly exceed available connection capacity, engineering bandwidth and system-planning certainty. Under these conditions the quality question moves upstream. Connection capacity, protection philosophy, reinforcement requirements, grid-impact study results, network charges, power-quality constraints and operating obligations become early selection criteria rather than late delivery concerns.

The Reifegradverfahren should be read as more than an administrative allocation mechanism. It makes visible that grid connectability has itself become a systems-engineering question. It also exposes a market reality that pipeline data make precise: a wide gap between connection demand expressed in requests and capacity actually delivered into operation. Data centres, large-scale renewables and new industrial loads are not isolated connection categories. Together they shape grid topology, flexibility needs, location quality and security of supply. Steering these fields separately while they interact technically generates additional complexity. Reading them together allows connection scarcity, project maturity and system value to become part of one infrastructure decision.

What governability means under operating conditions

The central category of this brief is governability. It denotes the capacity to keep a giga-scale data-centre campus, across its lifetime, in technically, evidentially and institutionally legible states: through active stewardship of the admissible operating envelope, coordination of electrical, thermal, connectivity and safety layers, consistent evidence across operation and change, governance of contractual interfaces, and continued relevance under changing regulatory, market and technical conditions. Governability is not an abstract aspiration. It determines whether a technically capable campus actually becomes bankable, operable and defensible infrastructure.

This matters because a giga-scale campus now operates within a system shaped simultaneously by AI density escalation, grid-connection scarcity, heat-planning obligations, carbon-reporting requirements, cybersecurity regulation and community expectations. A campus can hold nominal capacity and still be grid-fragile. It can carry planning consent and still lack evidence integrity. It can claim AI readiness and still be unable to deliver the cooling, power and commissioning conditions the workload requires. These tensions cannot be resolved by capacity announcements or technology optimism alone. They require an integrated view of system role, grid connection, operating envelope, evidence and governance.

Five disciplines become decisive.

First, treat the campus as a cyber-physical utility infrastructure object rather than as a building with IT load. Site, electrical architecture, cooling, connectivity, security, evidence, sustainability and governance layers are analytically separable and operationally inseparable. A weakness in one layer impairs the value of the others; coherence across all of them is the condition under which the asset becomes durable.

Second, treat grid connection as the first operating condition, not the last procurement step. Connection point, protection philosophy, reinforcement requirements and operating obligations shape what the campus can deliver, adapt and defend over time. A connection treated as an administrative task hardens design choices that later prove costly to revise — and under maturity-based allocation, weakens the project's ability to be argued credibly at the transmission system operator.

Third, govern the physical form of AI capacity as an active design discipline. Rack-level power, liquid-cooling pathways, CDU placement, network fabric, floor loading and commissioning logic must be treated as coupled design parameters rather than sequential upgrades. AI readiness is a system condition set at design stage, not a product specification added after the building shell has been fixed.

Fourth, treat evidence as an operating asset. Design models, commissioning records, configuration data, energy and water accounts, carbon allocation, maintenance records and customer reporting must remain part of one governed evidence architecture across the asset's life. Without that continuity, performance claims lose auditability precisely when they matter most — in refinancing, customer due diligence, regulatory assessment or incident reconstruction.

Fifth, align technical architecture with governance architecture. In multi-party campus configurations — developer, EPCM contractor, asset owner, operator, customer, grid operator — ownership boundaries, design authority, change-control logic and liability allocation shape whether technically sound decisions can actually be made and defended. Governance architecture is part of operability, not a contractual afterthought.

What Europe should preserve under scale-up

Europe does not primarily face a lack of technical insight in this field. Grid operators, developers, investors and technology providers have long understood that a power system shaped by high renewable penetration, large new industrial loads and hyperscale compute demand follows a different operating logic from one historically organised around a stable base of large thermal generators. The data centre is a load centre, a flexibility asset, a heat source, a regulatory object and a community partner simultaneously. Treating it as only one of these misses the infrastructure challenge.

The deeper tension lies in the incomplete institutional translation of that understanding into stable long-term decision architecture. When connection regimes, permitting frameworks, heat-planning obligations and evidence requirements are renegotiated within short political cycles, the result is not a neutral pause. It is a risk field in which projects become more expensive, capital becomes more cautious and system integration becomes more conflict-prone later. For giga-scale data centres this is especially consequential, because technical and operational quality must be formed early — in site selection, connection design, cooling architecture and evidence structure — while systemic value often becomes visible only during later operation.

Closing perspective

A giga-scale data centre should be read as more than a building, a power request, a colocation product or a nominal IT load. It is an infrastructure unit whose long-term value depends not only on power rating, commissioned capacity or revenue potential, but on whether it remains governable across its real lifetime: connectable, coolable, evidenced, regulation-compliant, system-relevant and economically operable under the conditions that will actually prevail. That is the standard against which the next phase of European data-centre development should be judged.